Minimizing Contractor Reliance & Credit Purchases to Strengthen Direct Customer Relationships and Improve Cashflow in a Plywood and Interior Materials Business

A Proposal Report for the BDM Capstone Project

Name: **Vaibhav Satish**

Roll Number: **21f3002068**

21f3002068@ds.study.iitm.ac.in

IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project Title "Minimizing Contractor Reliance & Credit Purchases to Strengthen Direct Customer Relationships in a Plywood and Interior Materials Business". I extend my appreciation to S.K. Timber, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.
I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.
I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

V.Satish

Signature of Candidate: (Digital Signature)
Name: Vaibhav Satish
Date: 12/05/2025

2. Executive Summary

SK Timber, a leading plywood and interior materials supplier in Aligarh since 2000, specializes in high-quality construction materials for residential and commercial projects. With a network of 600+ contractors, the business thrives on its reputation for fair pricing and industry expertise but faces critical operational challenges.

The company struggles with overdependence on contractors (80% of sales), hidden credit risks (discrepancy between reported and actual credit exposure), and seasonal demand volatility, which disrupts cash flow and inventory planning. These issues erode pricing power, strain financial visibility, and limit growth potential.

This project will employ a mixed-method approach, combining quantitative analysis (sales data, credit audits, inventory trends) with qualitative insights (contractor interviews). Tools like ABC segmentation, What-If Analysis, and EOQ modeling will identify strategies to:

1. Diversify sales channels by strengthening direct customer relationships,
2. Reform credit policies to reduce unrecorded liabilities, and
3. Align inventory with seasonal demand cycles to minimize stock imbalances.

The expected outcomes include improved profit margins, stabilized cash flow, and data-driven decision-making, enabling SK Timber to achieve sustainable scalability. By addressing these inefficiencies, the business can enhance competitiveness in Aligarh's growing modular interiors market.

3. Organization Background

S.K. Timber is a well-established plywood and interior materials business located at Eta Chungi, Near Medha Hospital, Aligarh, Uttar Pradesh (202001). Founded in 2000 by Mohammad Aslam, the company began as a small furniture workshop specializing in beds, sofas, and dressing tables. However, due to cash flow challenges caused by customers' credit-based purchases (primarily newlyweds), Aslam pivoted to the plywood trade, recognizing its cash-driven supplier model as more sustainable.

Over 25 years, S.K. Timber has grown into a trusted supplier of premium construction materials, including plywood, laminates, modular kitchens, doors, and hardware. With a product range of over 300 items, the business caters to both residential and commercial projects, serving contractors, carpenters, and end customers across Aligarh. Operating from a single location, it is supported by a team of 10 employees and maintains strong ties with 10–12 reliable suppliers. Its competitive edge lies in quality products, fair pricing, and deep industry expertise. Today, S.K. Timber leverages a network of 600+ contractors and a data-driven approach to stay ahead of trends, particularly the rising demand for modular interiors and high-end finishes.

4. Problem Statement

4.1 Contractor Over-Reliance Threatening Direct Customer Relationship

Explore contractor resale patterns and recommend strategies to diversify channels and strengthen direct customer relationships.

4.2 Hidden Credit Risk Exposure Limiting The Cashflow

Analyze payment behavior and existing credit practices to uncover the true credit exposure and suggest data-backed credit management processes.

4.3 Seasonal Demand Volatility Disrupting Inventory Planning

Identify seasonality patterns and assess inventory turnover rates to propose stock optimization strategies aligned with demand cycles.

5. Background of the problem

SK Timber, operating in Aligarh for over two decades, has established a strong presence in the plywood and interior materials market. Despite a wide contractor network of over 600, the business faces structural inefficiencies that limit scalability and long-term sustainability. A key concern is its overdependence on contractors, who contribute over 80% of sales. This limits direct customer engagement, weakens pricing power, and restricts access to valuable market insights, affecting brand control and loyalty. Additionally, while only 5% of sales are officially recorded as credit, nearly 10% of daily operational time is spent on payment recovery. This suggests the presence of informal or undocumented credit practices, which obscure the company's true financial position and hinder cash flow planning. Finally, SK Timber struggles with seasonal demand fluctuations—particularly during winter (January), monsoons (July-August), and agricultural cycles. These fluctuations disrupt inventory turnover and working capital management. In the absence of a structured, data-driven stocking strategy, the company risks frequent overstocking or understocking, further impacting profitability and operational efficiency.

Addressing these challenges through systematic analysis of sales, credit, and inventory data will help the business improve visibility, reduce inefficiencies, and prepare for sustainable growth in Aligarh's evolving interiors market.

6. Problem-Solving Approach

6.1 Method Use

A mixed-method approach will be used to explore and address the structural challenges faced by SK Timber. This approach blends quantitative data analysis with qualitative insights to ensure that both operational metrics and on-ground realities are captured effectively. The research will follow a three-pronged diagnostic model:

- **Channel Dependence Analysis** to evaluate the overreliance on contractors and its implications for pricing power and customer ownership.
- **Credit Risk Mapping** to uncover the extent and operational impact of undocumented credit.
- **Demand-Inventory Synchronization** to align procurement and stock levels with seasonal patterns.

This structured investigation will enable actionable, data-backed interventions that improve efficiency, customer engagement, and financial visibility.

6.2 Intended Data Collection

Data will be collected across three domains:

- **Sales & Channel Data**: Sales records for the past 12–18 months will be collected and classified by customer type (contractor vs. direct). Contractor markups will be analyzed to quantify the loss in pricing power and understand margins influenced by intermediary sales. Additional details such as product mix, ticket size, and frequency will be used to identify over-dependence and untapped customer segments.
- **Credit & Recovery Records**: Ledger books, payment receipts, and effort logs (formal and informal) will be examined to trace actual credit exposure. Interviews with the owner and collection staff will surface hidden liabilities and quantify operational inefficiencies.
- **Inventory & Seasonality Patterns**: Monthly stock-in and stock-out records will be analyzed alongside seasonal demand cues (weather, festivals, agricultural cycles) to identify mismatches between inventory planning and actual sales flow.

Complementing this will be **interviews with contractors and end customers** offering behavioral and contextual inputs that numbers alone may miss.

6.3 Analysis Tools

The study will leverage:

- Descriptive Analytics (MS Excel/Google Sheets/Power BI) for sales trends and credit aging patterns
- Correlation Analysis (Python/R) to link agricultural cycles with demand shifts
- Customer Segmentation (RFM or ABC Analysis) to classify contractors by sales volume/recency
- What-If Analysis (Excel's Scenario Manager) to model the impacts of changing key variables.

7. Expected Timeline

7.1 Work Breakdown Structure

7.1.1 Business Onboarding and Initial Engagement

7.1.1.1 Finalize business prospect and establish first contact
7.1.1.2 Prepare Questionnaire and conduct the first owner interview

7.1.2 Problem Identification and Setting Objectives

7.1.2.1 Create transcript, identify bottlenecks, and structure problem statements

7.1.2.2 Conduct follow-up discussion and finalize project objectives and data requirements

7.1.3 Data Collection

7.1.3.1 Collect access to sales, credit, and inventory data

7.1.3.2 Review ledgers, recovery logs, and conduct 1-2 stakeholder interviews

7.1.4 Data Cleaning and Analysis

7.1.4.1 Clean, correct, and model data for descriptive insights

7.1.4.2 Discussing Initial Findings with Business Owner

7.1.5 Deeper Research and Analysis

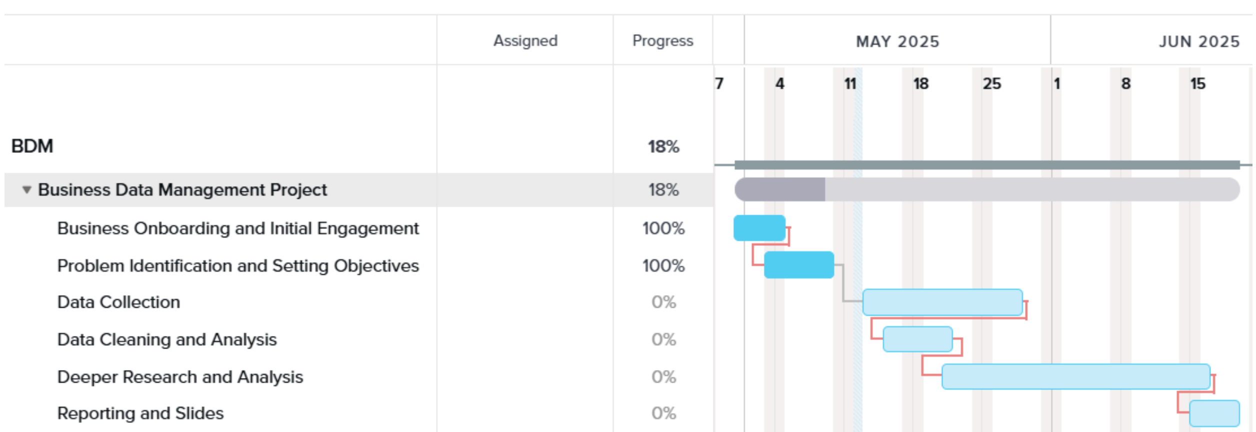
7.1.5.1 Segment customers/contractors (ABC/RFM), run What-If analysis

7.1.5.2 Develop EOQ-based inventory model and detect seasonal patterns

7.1.6 Reporting and Slides

7.1.6.1 Draft actionable recommendations and create final presentation slides

7.2 Gantt Chart



8. Expected Outcome

This project aims to uncover data-driven insights that illuminate the root causes and patterns behind SK Timber’s operational challenges—contractor dependence, hidden credit exposure, and seasonal demand volatility.

Reduced Contractor Overdependence: Through the analysis of sales and channel data, the study is expected to identify the degree of revenue concentration among contractors, variation in sales performance across customer types, and the presence of under-leveraged customer segments. Contractor-level contribution patterns, markup values, and sales frequency trends will offer a clearer picture of SK Timber’s dependency dynamics.

Improved Credit Management: By auditing credit records and payment histories, the project will quantify the actual credit exposure, beyond reported figures, and reveal operational inefficiencies in receivables management, including delays and informal credit extensions.

Optimized Inventory for Seasonality: Inventory and sales data, when examined alongside seasonal and agricultural cycles, will help surface recurring demand patterns, stock turnover inconsistencies, and periods of over- or under-supply.

Collectively, these insights will provide a fact-based foundation for developing strategies to enhance pricing control, cash flow stability, and inventory efficiency in the final phase of the project.